

## Lecture - 3

CO4: Compare various statistical methods of study data samples

CO5: Analyze and evaluation of different sets of data using hypothesis testing.

### JOINT, MARGINAL AND CONDITIONAL PROBABILITY MASS FUNCTIONS

Let us again consider the example.

As an illustration, let us consider the following example:

Let three balls  $b_1$ ,  $b_2$ ,  $b_3$  be placed randomly in three cells. The possible outcomes of placing the three balls in three cells are shown in Table.

| Arrangement<br>No. | Placement of the Balls in |            |        |
|--------------------|---------------------------|------------|--------|
|                    | Cell 1                    | Cell 2     | Cell 3 |
| 1                  | $b_1$                     | $b_2$      | $b_3$  |
| 2                  | $b_1$                     | $b_3$      | $b_2$  |
| 3                  | $b_2$                     | $b_1$      | $b_3$  |
| 4                  | $b_2$                     | $b_3$      | $b_1$  |
| 5                  | $b_3$                     | $b_1$      | $b_2$  |
| 6                  | $b_3$                     | $b_2$      | $b_1$  |
| 7                  | $b_1, b_2$                | $b_3$      | -      |
| 8                  | $b_1, b_2$                | -          | $b_3$  |
| 9                  | -                         | $b_1, b_2$ | $b_3$  |
| 10                 | $b_1, b_3$                | $b_2$      | -      |
| 11                 | $b_1, b_3$                | -          | $b_2$  |
| 12                 | -                         | $b_1, b_3$ | $b_2$  |
| 13                 | $b_2, b_3$                | $b_1$      | -      |
| 14                 | $b_2, b_3$                | -          | $b_1$  |

|    |                 |                 |                 |
|----|-----------------|-----------------|-----------------|
| 15 | -               | $b_2, b_3$      | $b_1$           |
| 16 | $b_1$           | $b_2, b_3$      | -               |
| 17 | $b_1$           | -               | $b_2, b_3$      |
| 18 | -               | $b_1$           | $b_2, b_3$      |
| 19 | $b_2$           | $b_3, b_1$      | -               |
| 20 | $b_2$           | -               | $b_3, b_1$      |
| 21 | -               | $b_2$           | $b_3, b_1$      |
| 22 | $b_3$           | $b_1, b_2$      |                 |
| 23 | $b_3$           | -               | $b_1, b_2$      |
| 24 | -               | $b_3$           | $b_1, b_2$      |
| 25 | $b_1, b_2, b_3$ | -               | -               |
| 26 | -               | $b_1, b_2, b_3$ | -               |
| 27 | -               | -               | $b_1, b_2, b_3$ |

**Table: Possible Outcomes of Placing the Three Balls in Three Cells**

Now, let  $X$  denote the number of balls in Cell 1 and  $Y$  be the number of cells occupied. Notice that  $X$  and  $Y$  are discrete random variables where  $X$  take on the values 0, 1, 2, 3 (number of balls in Cell 1 may be 0 or 1 or 2 or 3) and  $Y$  take on the values 1, 2, 3 (number of occupied cells may be 1 or 2 or 3). The possible values of two-dimensional random variable  $(X, Y)$ , therefore, are all ordered pairs of the values  $x$  and  $y$  of  $X$  and  $Y$ , respectively, i.e. are (0, 1), (0, 2), (0, 3), (1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (2, 3), (3, 1), (3, 2), (3, 3). Now, to each possible value  $(x_i, y_j)$  of  $(X, Y)$ , we can associate a number  $p(x_i, y_j)$  representing  $P(X = x_i, Y = y_j)$  as discussed.

In this example, we have obtained all possible values of  $(X, Y)$ , where  $X$  is the number of balls in Cell 1 and  $Y$  be the number of occupied cells. Now, let us associate numbers  $p(x_i, y_j)$  representing  $P(X = x_i, Y = y_j)$  as follows:

|                                                                                                                                                    |
|----------------------------------------------------------------------------------------------------------------------------------------------------|
| $p(0, 1) = P[X=0, Y=1] = P[\text{no ball in cell 1 and 1 cell occupied}]$<br>$= P[\text{Arrangement numbers 26, 27}] = 2/27$                       |
| $p(0, 2) = P[X=0, Y=2] = P[\text{no ball in cell 1 and 2 cell occupied}]$<br>$= P[\text{Arrangement numbers 9, 12, 15, 18, 21, 24}]$<br>$= 6/27$   |
| $p(0, 3) = P[X=0, Y=3] = P[\text{no ball in cell 1 and 3 cell occupied}]$<br>$= P[\text{impossible event}] = 0$                                    |
| $p(1, 1) = P[X=1, Y=1] = P[\text{one ball in cell 1 and 1 cell occupied}]$<br>$= P[\text{impossible event}] = 0$                                   |
| $p(1, 2) = P[X=1, Y=2] = P[\text{one ball in cell 1 and 2 cell occupied}]$<br>$= P[\text{Arrangement numbers 16, 17, 19, 20, 22, 23}]$<br>$= 6/27$ |
| $p(1, 3) = P[X=1, Y=3] = P[\text{one ball in cell 1 and 3 cell occupied}]$<br>$= P[\text{Arrangement numbers 1 to 6}] = 6/27$                      |
| $p(2, 1) = P[X=2, Y=1] = P[\text{two ball in cell 1 and 2 cell occupied}]$<br>$= P[\text{impossible event}] = 0$                                   |
| $p(2, 2) = P[X=2, Y=2] = P[\text{two ball in cell 1 and 2 cell occupied}]$<br>$= P[\text{Arrangement numbers 7, 8, 10, 11, 13, 14}]$<br>$= 6/27$   |
| $p(2, 3) = P[X=2, Y=3] = P[\text{two ball in cell 1 and 3 cell occupied}]$<br>$= P[\text{impossible event}] = 0$                                   |
| $p(3, 1) = P[X=3, Y=1] = P[\text{three ball in cell 1 and 2 cell occupied}]$<br>$= P[\text{Arrangement number 25}] = 1/27$                         |
| $p(3, 2) = P[X=3, Y=2] = P[\text{three ball in cell 1 and 2 cell occupied}]$<br>$= P[\text{impossible event}] = 0$                                 |
| $p(3, 3) = P[X=3, Y=3] = P[\text{three ball in cell 1 and 3 cell occupied}]$<br>$= P[\text{impossible event}] = 0$                                 |

The values of (X, Y) together with the number associated as above constitute what is known as joint probability distribution of (X, Y) which can be written in the tabular form also as shown below:

|       | Y | 1    | 2     | 3    | Total |
|-------|---|------|-------|------|-------|
| X     |   |      |       |      |       |
| 0     |   | 2/27 | 6/27  | 0    | 8/27  |
| 1     |   | 0    | 6/27  | 6/27 | 12/27 |
| 2     |   | 0    | 6/27  | 0    | 6/27  |
| 3     |   | 1/27 | 0     | 0    | 1/27  |
| Total |   | 3/27 | 18/27 | 6/27 | 1     |
|       |   |      |       |      |       |

We are now in a position to define joint, marginal and conditional probability mass functions

### Joint Probability Mass Function

Let  $(X, Y)$  be a two-dimensional discrete random variable. With each possible outcome  $(x_i, y_i)$ , we associate a number  $p(x_i, y_i)$  representing

$P[X = x_i, Y = y_i]$  or  $P[X = x_i \cap Y = y_i]$  and satisfying following conditions:

- i.  $p(x_i, y_i) \geq 0$
- ii.  $\sum_i \sum_j p(x_i, y_i) = 1$

The function  $p$  defined for all  $(x_i, y_i)$  is in analogy with one-dimensional case and called the **joint probability mass function of X and Y**. It is usually represented in the form of the table as shown in the example discussed above.

### Marginal Probability Function

Let  $(X, Y)$  be a discrete two-dimensional random variable which take up finite or countably infinite values  $(x_i, y_i)$ . For each such two-dimensional random

variable  $(X, Y)$ , we may be interested in the probability distribution of  $X$  or the probability distribution of  $Y$ , individually.

Let us again consider the example of the random placement of three balls in three cells wherein  $X$  and  $Y$  are the discrete random variables representing “the number of balls in Cell 1” and “the number of occupied cells”, respectively. Let us consider Table showing the joint distribution of  $(X, Y)$ . From this table, let us take up the row totals and column totals. The row totals in the table represent the probability distribution of  $X$  and the column totals represent the probability distribution of  $Y$ , individually. That is,

$$P[X = 0] = \frac{2}{27} + \frac{6}{27} + 0 = \frac{8}{27}$$

$$P[X = 1] = 0 + \frac{6}{27} + \frac{6}{27} = \frac{12}{27}$$

$$P[X = 2] = 0 + \frac{6}{27} + 0 = \frac{6}{27}$$

$$P[X = 3] = \frac{1}{27} + 0 + 0 = \frac{1}{27}$$

$$P[Y = 1] = \frac{2}{27} + 0 + 0 + \frac{1}{27} = \frac{3}{27}$$

$$P[Y = 2] = \frac{6}{27} + \frac{6}{27} + \frac{6}{27} + 0 = \frac{18}{27}$$

$$P[Y = 3] = 0 + \frac{6}{27} + 0 = \frac{6}{27}$$

These distributions of X and Y, individually, are called the marginal probability distributions of X and Y, respectively.

So, if (X, Y) is a discrete two-dimensional random variable which take up the values  $(x_i, y_i)$ , then the probability distribution of X is determined as follows:

$$p(x_i) = P[X = x_i]$$

$$p(x_i) = P[(X = x_i \cap Y = y_1) \text{ or } (X = x_i \cap Y = y_2) \text{ or } \dots]$$

$$p(x_i) = P[(X = x_i \cap Y = y_1) + (X = x_i \cap Y = y_2) + (X = x_i \cap Y = y_3) + \dots]$$

$$p(x_i) = \sum_j P[X = x_i \cap Y = y_j]$$

$$p(x_i) = \sum_j p(X = x_i) \quad [\because p(x_i, y_i), \text{ the joint probability mass function is } p[X = x_i \cap Y = y_i]]$$

which is known as the marginal probability mass function of X. Similarly, the probability distribution of Y is

$$p(y_j) = P[Y = y_j]$$

$$p(y_j) = P[(X = x_1 \cap Y = y_j) \text{ or } (X = x_2 \cap Y = y_j) \text{ or } \dots]$$

$$p(y_j) = \sum_i P[X = x_i \cap Y = y_j]$$

$$p(y_j) = \sum_i p(x_i, y_j)$$

and is known as the marginal probability mass function of Y.

## Conditional Probability Mass Function

Let  $(X, Y)$  be a discrete two-dimensional random variable. Then the conditional probability mass function of  $X$ , given  $Y = y$  is defined as

$$p(x|y) = P(X = x|Y = y)$$

$$= \frac{P[X=x \cap Y=y]}{P[Y=y]}, \text{ provided } P[Y = y] \neq 0$$

$$\left[ \because P[A|B] = \frac{P[A \cap B]}{P[B]}, P(B) \neq 0 \right]$$

Similarly, the conditional probability mass function of  $Y$ , given  $X = x$ , is defined as

$$p(y|x) = P[Y = y|X = x] = \frac{P[Y=y \cap X=x]}{P[X=x]},$$

Let us again consider the example as already discussed in this section.

Suppose, we are interested in finding the conditional probability mass function of  $X$  given  $Y = 2$ . Then the conditional probabilities are found separately for each value of  $X$  given  $Y = 2$ . That is, we proceed as follows:

$$P[X = 0|Y = 2] = \frac{P[X = 0 \cap Y = 2]}{P[Y = 2]} = \frac{\frac{6}{27}}{\frac{18}{27}} = \frac{1}{3}$$

$$P[X = 1|Y = 2] = \frac{P[X = 1 \cap Y = 2]}{P[Y = 2]} = \frac{\frac{6}{27}}{\frac{18}{27}} = \frac{1}{3}$$

$$P[X = 2|Y = 2] = \frac{P[X = 2 \cap Y = 2]}{P[Y = 2]} = \frac{\frac{6}{27}}{\frac{18}{27}} = \frac{1}{3}$$

$$P[X = 3|Y = 2] = \frac{P[X = 3 \cap Y = 2]}{P[Y = 2]} = \frac{\frac{0}{27}}{\frac{18}{27}} = 0$$

[Note that values of numerator and denominator in the above expressions have already been obtained while discussing the joint and marginal probability mass functions in this section of the unit.]

## Independence of Random Variables

Two discrete random variables X and Y are said to be independent if and only if

$$P[X = x_i \cap Y = y_j] = P[X = x_i]P[Y = y_j]$$

[ $\because$  two events A and B are independent if and only if  $P(A \cap B) = P(A)P(B)$ ]

**Example 1:** The following table represents the joint probability distribution of the discrete random variable (X, Y):

|   | Y | 1   | 2   |
|---|---|-----|-----|
| X |   |     |     |
| 1 |   | 0.1 | 0.2 |
| 2 |   | 0.1 | 0.3 |
| 3 |   | 0.2 | 0.1 |



Find:

- i) The marginal distributions.
- ii) The conditional distribution of X given  $Y = 1$ .
- iii)  $P[(X + Y) < 4]$ .

**Solution:**

i) To find the marginal distributions, we have to find the marginal totals, i.e. row totals and column totals as shown in the following table:

| Y      | 1   | 2   | $p(x)$<br>(Totals) |
|--------|-----|-----|--------------------|
| X      |     |     |                    |
| 1      | 0.1 | 0.2 | 0.3                |
| 2      | 0.1 | 0.3 | 0.4                |
| 3      | 0.2 | 0.1 | 0.3                |
| $p(y)$ | 0.4 | 0.6 | 1                  |
| Totals |     |     |                    |

Thus, the marginal probability distribution of X is

| X      | 1   | 2   | 3   |
|--------|-----|-----|-----|
| $p(x)$ | 0.3 | 0.4 | 0.3 |

Thus, the marginal probability distribution of Y is

|      |     |     |
|------|-----|-----|
| Y    | 1   | 2   |
| p(y) | 0.4 | 0.6 |

ii)

$$P[X = 1|Y = 1] = \frac{P[X = 1, Y = 1]}{P[Y = 1]} = \frac{0.1}{0.4} = \frac{1}{4}$$

$$P[X = 2|Y = 1] = \frac{P[X = 2, Y = 1]}{P[Y = 1]} = \frac{0.1}{0.4} = \frac{1}{4}$$

$$P[X = 3|Y = 1] = \frac{P[X = 3, Y = 1]}{P[Y = 1]} = \frac{0.2}{0.4} = \frac{1}{2}$$

∴ The conditional distribution of X given Y = 1 is

|            |     |     |     |
|------------|-----|-----|-----|
| X          | 1   | 2   | 3   |
| P[X=x Y=y] | 1/4 | 1/4 | 1/2 |

iii) As the values of (X, Y) which satisfy  $X + Y < 4$  are (1, 1), (1, 2) and (2, 1) only.

$$\therefore P[X + Y < 4] = P[X = 1, Y = 1] + P[X = 1, Y = 2] + P[X = 2, Y = 1]$$

$$= 0.1 + 0.2 + 0.1 = 0.4$$

**Example 2:** Two discrete random variables X and Y have

$P[X = 0, Y = 0] = 2/9$ ,  $P[X = 0, Y = 1] = 1/9$ ,  $P[X = 1, Y = 0] = 1/9$ , and  $P[X = 1, Y = 1] = 5/9$ . Examine whether X and Y are independent?

**Solution:** Writing the given distribution in tabular form as follows:

|      | Y | 0   | 1   | p(x) |
|------|---|-----|-----|------|
| X    |   |     |     |      |
| 0    |   | 2/9 | 1/9 | 3/9  |
| 1    |   | 1/9 | 5/9 | 6/9  |
| p(y) |   | 3/9 | 6/9 | 1    |

$$\therefore P[X = 0] = \frac{3}{9}, P[X = 1] = \frac{6}{9}$$

$$P[Y = 0] = \frac{3}{9}, P[Y = 1] = \frac{6}{9}$$

$$\text{Now } P[X = 0] P[Y = 0] = \frac{3}{9} * \frac{3}{9} = \frac{1}{9}$$

$$\text{But } P[X = 0, Y = 0] = \frac{2}{9}$$

$$\therefore P[X = 0, Y = 0] \neq P[X = 0] P[Y = 0]$$

Hence X and Y are not independent.

[Note: If  $P[X = x, Y = y] = P[X = x] P[Y = y]$  for each possible value of  $X$  and  $Y$ , only then  $X$  and  $Y$  are independent.]]

Here are two exercises for you.

E1) The joint probability distribution of a pair of random variables is given by the following table:

| Y | 1    | 2   | 3    |
|---|------|-----|------|
| X |      |     |      |
| 1 | 1/12 | 0   | 1/18 |
| 2 | 1/6  | 1/9 | 1/4  |
| 3 | 0    | 1/5 | 2/15 |

i) Evaluate marginal distribution of  $X$ .

ii) Evaluate conditional distribution of  $Y$  given  $X = 2$

iii) Obtain  $P[X + Y < 5]$ .

E2) For the joint probability distribution of  $(X, Y)$ ,

| Y | 1    | 2    | 3    |
|---|------|------|------|
| X |      |      |      |
| 1 | 1/20 | 1/10 | 1/10 |

|   |        |        |        |
|---|--------|--------|--------|
| 2 | $1/20$ | $1/10$ | $1/10$ |
| 3 | $1/10$ | $1/10$ | $1/20$ |
| 4 | $1/10$ | $1/10$ | $1/20$ |

i) find the probability that  $Y = 2$  given that  $X = 4$ ,

ii) find the probability that  $Y = 2$ , and

iii) examine if the two events  $X = 4$  and  $Y = 2$  are independent.